Olist E-Commerce

Delivery Performance & SLA Analysis

Are we meeting our delivery SLAs,
and how does delivery performance impact customer satisfaction?

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Brazilian E-Commerce Dataset — Olist | 2017–2018

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1. Executive Summary

Olist processed 96,470 delivered orders between 2017 and 2018. While the platform maintains an overall on-time delivery rate of 93.22%, a critical 6.77% late delivery rate is directly damaging customer satisfaction — orders delivered late receive a review score of 2.3/5, compared to 4.3/5 for on-time orders. The root cause is not sellers or order processing; it is the carrier shipping stage, which accounts for 90.2% of all delays.

93.22%	6.77%	96,470	4.3 vs 2.3
On-Time Delivery Rate	Late Delivery Rate	Total Delivered Orders	Satisfaction: On-Time vs Late

Key Finding:

90.2% of delayed orders are caused by the shipping carrier stage — not by sellers or order processing. Northern Brazilian states (AL, MA, SE, PI) account for the highest late delivery rates, driven by geographic distance from São Paulo where most sellers are located.

2. SLA Overview — Key Performance Indicators

A Service Level Agreement (SLA) in delivery is the formal commitment that an order will be delivered on or before the estimated delivery date. The following KPIs measure how well Olist is honoring that commitment.

2.1 Overall Delivery Status

Metric	Value	Status
Total Delivered Orders	96,470	—
On-Time Orders	89,936	**✓ Target Met**
Late Orders	6,534	**⚠ Needs Attention**
On-Time Delivery Rate	93.22%	✓ Acceptable
Late Delivery Rate	6.77%	**⚠ Risk Level**

2.2 Delay Severity Breakdown

When delivery delays occur, categorizing them by severity reveals that while most delays are minor, a meaningful 5.5% of all late orders represent severe operational failures.

Delay Category	Avg. Delay (Days)	% of Late Orders	Business Risk
< 1 Week	3.1 days	48.81%	Low
1 Week – 1 Month	13.3 days	45.68%	Moderate
1 Month – 3 Months	43.3 days	4.79%	**High**
> 3 Months	129.4 days	0.72%	**Critical**
Severe Delays Total (> 1 Month)	—	**5.51%**	**⚠ SLA Breach**

Key Insight: 94.49% of delays are minor (under one month) → indicating the logistics network is functional but inconsistent. However, the 5.51% severe delay rate signals critical SLA breaches that risk customer churn and platform reputation damage.

3. Time Analysis

Analyzing late delivery rates across months and years reveals seasonal patterns and a concerning year-over-year deterioration in SLA performance.

3.1 Late Delivery Rate by Year

Year	Peak Month	Peak Late Rate	Lowest Month
2017	November	**12.5%**	June – August (~2.5%)
2018	March	**19.0%**	June (~1.2%)

Key Insight: The late delivery rate worsened significantly in early 2018 — peaking at 19% in March 2018 vs. 12.5% in November 2017 → suggesting the platform's logistics network failed to scale with the rapid growth in order volume, directly risking customer loyalty during the platform's most critical growth phase.

3.2 Order Volume vs. Late Rate

Analysis of monthly order volume against late delivery rates reveals an important finding: higher order volume does not directly cause higher late rates. The correlation is weak, indicating that the bottleneck is structural (geographic and carrier-related) rather than volume-driven.

4. Geographic Performance

Geographic analysis reveals a clear North–South divide in delivery reliability. Northern states, which are furthest from São Paulo where most sellers are concentrated, consistently record the highest late delivery rates.

4.1 Late Delivery Rate by State (Top 10 Worst)

Rank	State	Late Rate %	Region
1	AL — Alagoas	**21.4%**	Northeast
2	MA — Maranhão	**17.4%**	Northeast
3	SE — Sergipe	**15.2%**	Northeast
4	PI — Piauí	13.9%	Northeast
5	CE — Ceará	13.8%	Northeast
...	Best Performers: AM, RO, AP	~2.8%–3.8%	North/Amazon

4.2 Top 10 Cities by Late Rate

Rank	City	Late Rate %	Late Orders
1	Maceió	**27.1%**	64
2	São Gonçalo	**21.6%**	83
3	Cabo Frio	21.2%	24
4	Nova Friburgo	19.7%	28
5	São Luís	19.4%	65
6	Teresina	18.8%	51
7	Rio das Ostras	17.7%	23
8	São João de Meriti	16.1%	20
9	Macaé	15.8%	36
10	Resende	15.7%	18

Key Insight: The top late-delivery states are all in Brazil's Northeast — the region furthest from São Paulo's seller hub → confirming that geographic distance is the primary structural driver of delays. Resolving this requires regional distribution centers, not seller or process improvements.

5. Seller Impact

Seller analysis examines whether specific sellers are causing disproportionate delays, and whether high-volume sellers are more reliable than low-volume ones.

5.1 Seller Volume vs. Late Rate

Scatter plot analysis of all sellers (minimum order threshold applied) reveals a clear inverse relationship: high-volume sellers maintain lower late rates (5–10%), while small-volume sellers show higher variance, with some reaching late rates above 19%.

Seller Tier	Order Volume	Avg. Late Rate	Reliability
Top Sellers	> 1,000 orders	5% – 10%	**✓ More Reliable**
Mid-Tier Sellers	200 – 1,000 orders	5% – 14%	Moderate
Small Sellers	< 200 orders	**Up to 19%**	**⚠ High Variance**

Key Insight: Top sellers are significantly more reliable → suggesting that experience and operational scale correlate with better logistics coordination. However, even the best sellers are subject to carrier delays, confirming that seller performance alone does not determine on-time delivery.

6. Operational Bottlenecks

To identify where delays occur in the order lifecycle, the journey was divided into three stages: order processing, seller fulfillment, and carrier shipping. Each stage was measured for both on-time and late orders.

6.1 Order Journey Stage Analysis

Stage	Overall Avg.	On-Time Orders	Late Orders	Bottleneck?
Processing Time (Purchase → Approval)	0.26 days	0.26 days	0.33 days	No
Seller Fulfillment (Approval → Carrier)	−3.85 days	−4.06 days	−0.95 days	No
Shipping Time (Carrier → Customer)	8.84 days	7.51 days	**27.38 days**	**YES ⚠**

Difference in shipping time between on-time and late orders: +19.87 days — the entire gap between on-time and late delivery is in the carrier stage.

6.2 Root Cause Attribution

Stage	% of Late Orders Affected	Interpretation
Shipping (Carrier)	**90.2%**	Primary bottleneck — carrier is responsible
Processing	21.0%	Secondary — minor overlap with shipping delays
Seller Fulfillment	20.2%	Secondary — some sellers slower on late orders

Note: Percentages do not sum to 100% — a single late order can have delays across multiple stages simultaneously.

Key Insight: 90.2% of delays originate in the carrier shipping stage → the problem is not seller preparation or order processing. The solution lies in logistics infrastructure improvements, particularly for long-distance routes to northern Brazil.

6.3 Geographic Root Cause

States with the highest average shipping times are all in northern Brazil — far from the São Paulo seller hub:

State	Full Name	Avg. Shipping Time
RR	Roraima	**25.2 days**
AP	Amapá	**23.2 days**
AM	Amazonas	**23.1 days**

AL	Alagoas	**20.6 days**
PA	Pará	19.9 days
MA	Maranhão	17.5 days

7. Customer Satisfaction Impact

The business impact of delivery delays is most clearly reflected in customer review scores. The data reveals a dramatic and consistent relationship between delivery timing and customer satisfaction.

4.29 / 5 Avg. Score — On-Time Orders	**2.27 / 5** Avg. Score — Late Orders	**−2.02** Score Drop Due to Delay	**89,443 vs 6,381** Reviews: On-Time vs Late

Key Insight: Delivery delays cause a 47% drop in customer satisfaction scores (from 4.29 to 2.27 out of 5) → directly risking customer churn, negative reviews, and long-term revenue loss. This is not a minor operational inconvenience — it is a strategic business threat. Every 1% reduction in the late delivery rate protects approximately 965 customers from a poor experience.

The sample sizes are statistically robust: 89,443 reviews for on-time orders and 6,381 for late orders, confirming this is not an outlier effect.

8. Key Findings & Recommendations

8.1 Key Findings Summary

Finding	Business Impact
6.77% of orders are delivered late	Puts 6,534 customers at risk of churn per period
90.2% of delays originate in carrier shipping	Logistics infrastructure is the core problem, not sellers
Northern states have 21%+ late rates (AL, MA, SE)	Geographic inequality damages brand trust in key markets
Late delivery drops review score from 4.29 to 2.27	47% satisfaction drop threatens repeat purchase rates
5.51% of late orders are severe (>1 month delay)	Critical SLA breaches with high refund/complaint risk
Top sellers are more reliable (5–10% late rate)	Seller development programs can improve overall SLA

8.2 Strategic Recommendations

01 Invest in Northern Logistics Infrastructure

States AL, MA, SE, PI, and CE show late rates above 13.8%. Establish regional distribution centers or partner with local carriers in the Northeast to reduce shipping time from 20+ days to under 10 days. Impact: Estimated 40–60% reduction in late deliveries for these states.

02 Revise Estimated Delivery Dates for Remote Areas

Current estimated delivery dates for northern and northeastern states are unrealistic, setting expectations the carrier consistently fails to meet. Adjusting estimates to reflect actual shipping times will reduce the measured "late" rate and improve customer experience through accurate expectation-setting.

03 Carrier Performance Audit

Since 90.2% of delays are carrier-driven, identify which specific carriers are responsible for the worst-performing routes. Introduce carrier SLA penalties and consider competitive bidding for underperforming routes.

04 Seller Development for Small Sellers

Small sellers (< 200 orders) show late rates up to 19%. While carrier delays are the primary cause, seller fulfillment speed on late orders lags by 3.11 days vs. on-time orders. A seller onboarding program focused on fast fulfillment can reduce this secondary factor.

9. Appendix — Indicator Definitions

This appendix defines all analytical indicators used throughout this report, including calculation methodology for non-standard metrics.

Indicator	Definition	Calculation / Notes
Late Order	An order where the actual delivery date exceeds the estimated delivery date set at purchase time.	order_delivered_customer_date > order_estimated_delivery_date
On-Time Delivery Rate	Percentage of delivered orders that arrived on or before the estimated date.	(On-Time Orders / Total Delivered Orders) × 100
Late Delivery Rate	Percentage of delivered orders that arrived after the estimated date.	(Late Orders / Total Delivered Orders) × 100
Average Delivery Delay	Mean number of days by which late orders exceeded the estimated delivery date.	Mean of (actual_date − estimated_date) for late orders only. Calculated per delay category to handle high variance.
Severe Delay Rate	Percentage of late orders with delays exceeding one month — classified as critical SLA breaches.	(Orders delayed > 30 days / Total Late Orders) × 100. Threshold: 1 month chosen as the standard SLA breach indicator.
Processing Time	Time from customer purchase to seller notification/approval.	order_approved_at − order_purchase_timestamp
Seller Fulfillment Time	Time from seller receiving the order to handing off to the carrier. Negative values indicate sellers shipping ahead of schedule.	order_delivered_carrier_date − order_approved_at. A negative avg. (−3.85 days) means sellers ship early on average.
Shipping Time	Time from carrier pickup to customer delivery. Identified as the primary bottleneck.	order_delivered_customer_date − order_delivered_carrier_date
Review Score	Customer satisfaction rating on a 1–5 scale, submitted post-delivery via Olist platform.	Averaged by delivery status (Early / Late). Based on 89,443 on-time and 6,381 late order reviews.
Late Rate by State / City	Percentage of orders delivered to a given state or city that were classified as late.	(Late Orders in Region / Total Delivered Orders in Region) × 100. Minimum order threshold applied for statistical reliability.
SLA (Service Level Agreement)	Formal commitment that an order will be delivered on or before the estimated delivery date provided at checkout.	In the Olist context, SLA compliance = actual delivery ≤ estimated delivery. No explicit SLA contract exists in the public dataset; the estimated date is used as the SLA benchmark.

Data Source: Olist Brazilian E-Commerce Public Dataset (Kaggle). Analysis period: January 2017 – August 2018. Only orders with status "delivered" and a recorded actual delivery date were included in this analysis.

All analyses performed using Python (pandas, NumPy, Matplotlib, Seaborn).